



Recognizing salat activity using deep learning models via smartwatch sensors

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Received: 27 October 2023 / Accepted: 5 July 2024 / Published online: 6 August 2024
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Abstract

In this study, we focus on human activity recognition, particularly aiming to distinguish the activity of praying (salat) from other daily activities. To achieve this goal, we have created a new dataset named HAR-P (Human activity recognition for Praying), which includes eight different activities: walking, running, sitting, standing, walking upstairs, walking downstairs, typing with a keyboard, and praying (salat). The HAR-P dataset was collected from 50 male individuals, who wore smartwatches on their dominant wrists. We compare the activity classification performance using three state-of-the-art algorithms from the literature: Long Short-Term Memory, Convolutional Long Short-Term Memory, and Convolutional Neural Network—Long Short-Term Memory. To assess the influence of sensors, data from accelerometer, gyroscope, linear acceleration sensor, and magnetic field sensor were utilized. The impact of individual sensor data as well as combinations thereof was investigated. The highest classification accuracy within single sensor groups, reaching 95.7%, was achieved using the accelerometer data with the Convolutional Long Short-Term Memory method. Combining two sensor groups resulted in an increase in accuracy of up to 9%. The highest accuracy of 96.4% was obtained by utilizing three sensor groups together with the Convolutional Neural Network—Long Short-Term Memory method. Furthermore, the evaluation of sensor and model performance was conducted using the stratified *k*-fold cross-validation method with 5-folds. These findings contribute significantly to evaluating the performance of sensor combinations and different algorithms in activity classification. This study may provide an effective foundation for the automatic recognition and tracking of human activities and offer an applicable model, particularly for the recognition of religious practices such as praying.

Keywords Human activity recognition · Smartwatch sensor data · Salat recognition · Time series classification

Abbreviations

1D	One dimensional	ConvLSTM	Convolutional long short-term memory
2D	Two dimensional	downstairs	Walking downstairs
acc	Linear acceleration sensor	GPS	Global positioning system
acg	Accelerometer sensor	gyro	Gyroscope sensor
CD	Custom dataset	HAR	Human activity recognition
CNN	Convolutional neural network	HAR-P	Name of the new dataset (Human activity recognition for Praying)
CNN-LSTM	Convolutional neural network—long short-term memory	KC	Kinect camera
		LSTM	Long short-term memory
		magnet	Magnetic field sensor
		SES	Special environment setup
		typing	Typing with a keyboard
		upstairs	Walking upstairs

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1 Introduction

Human Activity Recognition (HAR) is the area of research that focuses on determining a particular action or motion performed by an individual by analyzing data collected from different sensing devices like video cameras [1, 2] and sensors [3, 4]. The progress of wearable technology and the increasing enthusiasm for HAR have laid the groundwork for revolutionary applications across multiple fields. In the realm of healthcare, Paraschiakos and colleagues [5] harnessed accelerometer sensor data to monitor the well-being of the elderly, while Mohammadian Rad and team [6] utilized wearable sensors to oversee Parkinson's disease. In the domain of human–computer interaction, Zhao et al. [7] introduced the MobiGesture app, an intelligent system for recognizing hand gestures that are attuned to mobility, and Zhang et al. [8] presented FingerPing, a sophisticated system capable of discerning a variety of intricate hand poses.

Research in the field of human activity recognition has brought about significant advancements to ease our daily lives. However, certain special rituals, such as the prayer (salat), have often been overlooked. Salat holds deep religious significance as one of Islam's Five Pillars, and devout Muslims observe it at specific times five times a day: Fajr (dawn), Dhuhr (noon), Asr (afternoon), Maghrib (sunset), and Isha (night). However, the demands of modern life and time constraints can make it challenging to perform the prayer on time. Therefore, leveraging the capabilities offered by technology to recognize and facilitate the tracking of the prayer ritual is crucial.

In this study, a specific focus has been placed on prayer recognition as a particular focal point in the field of human activity recognition. Distinguishing prayer worship from daily other activities constitutes the main objective of this study. Especially, it is crucial to differentiate and recognize prayer from other activities for individuals not to miss prayer times and to perform their worship regularly.

Existing studies in prayer recognition mainly rely on image processing techniques. However, there are also studies utilizing sensor data, often focusing on devices such as mobile phones and conducting research to determine the most suitable position for mobile phones. However, as mobile phones can be carried in different positions by end-users, a successful final framework has not been developed. Additionally, mobile phones create noise in sensor values as they move excessively in places like pockets or bags, which adversely affect success. To overcome this drawback, we utilized smartwatches in this study. Existing studies on prayer recognition have focused on identifying the prayer segments and ensuring the correct performance of the prayer movements. In contrast, we focused on

distinguishing prayer worship from daily other activities and investigated which sensor or sensor combination could best accomplish this distinction.

The prominent contributions of this study are as follows:

- A new human activity dataset has been compiled, encompassing eight different daily activities including the prayer. It is anticipated to be utilized in future studies across various domains.
- We conducted tests consisting of fifteen different combinations of four sensor types to determine which sensor type yields higher accuracy in activity recognition
- We evaluated the performance of state-of-the-art deep learning algorithms using raw sensor data, bypassing traditional filtering methods.
- We were able to distinguish prayer worship from other daily activities as a whole with a 96.4% accuracy rate, rather than focusing on prayer segments.
- We enabled end-users to achieve high-accuracy activity classification using a wearable device such as a smartwatch, whose usage has increased in recent times.

The results of this study will lay a foundation for future researchers and contribute to the development of new approaches for integrating prayer worship with technology. Particularly, the development of mobile applications that can be used through portable devices such as smartwatches to monitor and remind users of prayer worship will be a significant step in this field.

The rest of this paper is organized as follows. Section 2 introduces the related work about HAR and salat recognition. Section 3 describes the HAR-P dataset, data collecting, data labeling, data alignment and frequency matching. Methodology, short explanation of the deep learning algorithms, and experimental setup are presented in Sect. 4. Results are presented and discussed in Sect. 5. Finally, concluding remarks and future work are presented in Sect. 6.

2 Related work

Various machine learning and deep learning techniques have been employed to detect human activities [9]. Mekruksavanich, Jitpattanakul [10] introduced the multi-channel CNN-LSTM (Convolutional Neural Network—Long Short-Term Memory) network; they achieved an accuracy of 96.87% in the DHA dataset consisting of accelerometer data. Ascioğlu, Senol [11] produced a new dataset for HAR. They collected acceleration and gyroscope data from one master and four slave devices. They tested their dataset with CNN (Convolutional Neural Network), LSTM (Long Short-Term Memory) and

ConvLSTM (Convolutional LSTM) model and they reported that the ConvLSTM model performed better.

We came across about 20 academic studies on prayer (salat) recognition. Half of them are based on image processing (summarized in Table 1) and the other half are sensor based studies (summarized in Table 2). El-Hoseiny, Shaban [12] focuses on analyzing prayer gestures using an image processing technique that can aid beginners in learning the accurate way to pray. By capturing a side view

of a person through a camera, the foreground and background are initially distinguished using a masking method. Subsequently, contours are identified. The different components of the prayer posture are categorized by extracting characteristics related to the angles formed by the back, head, and feet. They achieved a success average of 93.5%. Jaafar et al. [13], they tracked prayer movements using two Kinect cameras. In another study, they studied the positioning effect of two Kinect cameras [14]. In another study,

Table 1 Summary of vision-based praying (salat) recognition

Title	Device	Device location	Method/Algorithm	Class Count	Dataset	Score	Limitations
Muslim prayer actions recognition [12]	C	Side view of the prayer	Geometric parameters of the prayer's body	Qiam, ToRuku, Ruku, FromRukuToQiam, Sujud, Jalsa	26 videos with varying prayer parameters	93.5%	CD, SES
An Investigation Of Motion Tracking For Solat Movement With Dual Sensor Approach [13]	KC	First kinect camera front of prayer, second kinect camera side of prayer, fixed position setup	Hidden Markov Model	Takbir, Ruku', I'tidal, Sujud, Sit	All the movements are tracked for 10 times	100%	CD, SES
Optimal Dual Cameras Setup for Motion Recognition in Salat Activity [14]	KC	Two kinect camera with four position setup	Hidden Markov Model	Takbir, Ruku', I'tidal, Sujud, Sit	five main movements and for each movement, 10 samples	100%	CD, SES
A New Approach in Islamic Learning: Performance Evaluation of Motion Recognition System for Salat Movement [15]	KC and PM	Two kinect camera with four position setup	Hidden Markov Model	Comparing error rate with traditional tutor-based methodology	25 participants	Error Rate: 2.08	CD, SES
Activity Monitoring of Islamic Prayer(Salat) Postures using Deep Learning [16]	S	–	CNN, YoloV3	Standing, Bowing, Prostrating, Sitting	849 images	85%	CD
Monitoring and Alarming Activity of Islamic Prayer (Salat) Posture Using Image Processing [27]	KC	Front and back of prayer mat with fixed position	Template matching method	Takbiratul Ihram, Ruku', Sujud, At-Tawarrok	–	90%	CD, SES
Salat Activity Recognition on Smartphones using Convolutional Network [28]	S	In front of prayer mat	CNN	Takbeer, Al-Qiyam, Ruku, Qiyam, Sujud, Julus, Tashahhud, Salam	200 videos (4,992 labeled images)	88%	SES
Istaqim: An Assistant Application to Correct Prayer for Arab Muslims [29]	S	–	YoloV5	Takbeer, Bowing (ruku), Rising from bowing, Prostration(sujud), and Tashahhud	Youtube videos, images from internet	–	It is assumed that the person's entire body is accurately depicted

Device: Camera (C), Kinect Camera (KC), Smartphone camera (S), Prayer Mat (PM)

Limitations: Custom Dataset (CD), Special Environment Setup (SES)

Table 2 Summary of Sensor-based praying (salat) recognition

Title	Sensor(s)	Device(s)	Device location	Algorithm	Class count	Dataset	Score	Limitations
Monitor Pilgrims: Prayer Activity Recognition using Wearable Sensors [18]	ECG, GSR	W	Chest, wrist	SVM	The first case, between congregational prayers and individual prayers, in the second case, between silent prayers and loud prayers	10 participants in an 8-day pilgrimage	78% for the first case, and 84% for the second case	Umrah trip, differences in bio-physiological responses
Monitoring Prayer Using Mobile Phone Accelerometer [19]	A	S	Upper back	NB, KNN, J48	Standing, bowing, prostrating, sitting	Thirty rounds; each round consists of eight stages	NB 91.84%, IB1 87.76%, J48 89.84%	Non-ergonomic device position
Prayer Activity Monitoring and Recognition Using Acceleration Features with Mobile Phone[20]	A	S	Chest, arm, hip and upper back	NB, KNN, J48	Standing, bowing, prostration and sitting	Six prayers, totalling 118 samples	NB 91.3%, J48 90.2%, KNN 88%	Conducted with limited participants
Aggregated activity recognition using smart devices [21]	A	S	Shirt pocket around the chest	NB, KNN, J48	Walking, running, standing, sitting, bowing, prostration	Ten subjects		Limited age range, limited participants
Salat activity recognition using smartphone triaxial accelerometer [22]	A	S	In front or back pocket	DTW	Qayyam, Rukku, Qauma, Sijda, Jalsa, Second Sijda	60 samples of Imam, total of 200 samples	82.00%	Activities for which there is no proper comparison
SARM: Salah Activities Recognition Model Based on Smartphone [24]	A	S	Right upper hand, left upper hand, front right trouser pocket, and front left trouser pocket	RF, KNN, NB, DT	Standing, standing to bowing, and standing to prostration	Ten subjects, six different features	RF 97%, KNN 94%, NB 71.6%, DT 95%	Variable factors affecting data integrity
Prayer Activity Recognition Using Smartphone [25]	A, G, M	S	Pockets or on the floor	SVM	Qayyam, Rukku down, Rukku up, Sijdah one, Jalsa, Sijdah two	8 volunteers, 18,090 samples	93,92%	Limited age range, limited participants
Neural Networks Based Prayer Monitoring and Recognition Framework [26]	A, G, H	SW	Wrist	CNN	2 Raqah, 3 Raqah, 4 Raqah	134 prayers	87%	Interpreting data from a limited perspective

Sensors: Accelerometer (A), electrocardiography (ECG), galvanic skin response (GSR), gyroscope(G), heartrate electrocardiogram(H), magnetometer (M)

Devices: Wearable sensors (W), Smartphone(S), Smartwatch (SW)

Algorithms: Support Vector Machine (SVM), Naïve Bayes(NB), K-Nearest Neighbor(KNN), J48 Decision Trees(J48), Decision tree (DT), Random forest (RF), Dynamic Time Warping (DTW)

they used a prayer mat with seven sensors next to two Kinect cameras. In their last study, they compared the error rate of the person praying with the conventional tutor-based and developed prototype method [15]. They used the Hidden Markov Model method in these four studies. Koubâa et al. [16] developed an AI-powered framework that guides worshipers in assessing the accuracy of their prayer postures. They collected 849 images showing different prayer situations from the internet and reported that they obtained an average of 85% precision as a result of their classification with the YoloV3 algorithm. Rahman et al. [17] introduced an Intelligent Salat Monitoring and Training System based on machine vision and image processing. They utilized the conventional approach of image processing, specifically template matching, using the images captured by the camera positioned within the custom setup constructed from PVC pipes. Their suggested methods will be able to recognize and match the pattern up to 90% and inform the user about the prayer performance. The literature reviews we have summarized so far are based on image processing and are not efficient and useful for the end user as they involve camera placement in the environment.

Sensor-based prayer activity recognition studies are summarized in Table 2. Muaremi et al. [18] observed differences in subjects' bio-physiological responses during prayer. They reported that they could distinguish between the congregational prayer and the individual prayer in the first case, and the silent and loud prayer in the second case with SVM algorithm. Al-Ghannam, Kanjo, & Al-Dossari classified the four parts of the prayer with the acceleration sensor data obtained from the mobile phone. In their initial research [19], they collected data by placing the mobile phone on the upper back. In their second study, they investigated the impact of device placement by placing it on various areas, including the chest, arms, hips, and upper back [20]. Eskaf et al. [21] performed classification studies in two second time periods using the acceleration sensor on basic activities, including praying. Data collection was carried out with a mobile phone placed around the chest pocket. They collected data from 10 people, 8 adults and 2 children. By performing the prayer parts in order, the completion of rak'ahs was determined and a warning was given about how many rak'ahs were prayed. Weka is used with C#. They reported that the KNN algorithm gave better results than j48 and Naïve Bayes. Ali et al. [22] made a classification based on the similarity measurement of the time series with the dynamic time warping (DTW) algorithm, with the acceleration sensor data they obtained while praying for four rakats from the phone they placed in the front and back pockets of the users. In the classification, they first extracted the pattern from the prayer offered by the imam and compared the prayers of other users with the

prayer pattern of the imam. In his thesis, Alobaid [23] tried to classify the five parts of the prayer by collecting acceleration sensor data from twenty people by mobile phone. After preprocessing on the raw data, he extracted ninety features from the time and frequency domains. He tested different classification algorithms and reported that the random forest was the most successful with 90% accuracy. He proposed a two-level classifier on mixing the "sitting" and "standing" parts in classification and achieved a 3% success rate increase. He also studied the effect of personal characteristics on classification success. Ahmad et al. [24] did a study to classify the three parts of the prayer with the acceleration sensor data they obtained from the mobile phone that they placed in 4 different parts of the body. They combined the features they obtained from the time domain into six sets and compared the effect. They performed tests with four classifiers in MATLAB and reached the highest success rate of 97% with Random Forest. They stated that since the phone could not be fixed in the trouser pockets of the volunteers, excessive noise occurred in the data and left-hand and right-handed sensor placements were more suitable. Topu et al. [25] divided the prayer into six parts and collected data from eight people with a smartphone in their trouser pocket. It achieved 93% success with the SVM algorithm. In their study, they used three sensors: acceleration, gyroscope and magnetic field. They have proven that it is possible to improve accuracy by using multiple sensors without using a single sensor. Unlike other prayer recognition studies Al-Selwi et al. [26] used a smartwatch for the first time. They proposed a deep learning-based time series data classification model using three sensors: heart rate electrocardiogram, accelerometer and gyroscope. In their studies where they used the CNN algorithm, they made 87% accuracy classification.

3 HAR-P dataset

The primary objective of this study is to detect the unique activity of "praying" from other daily activities using smartwatch sensor data. To facilitate this, we created a new dataset named "Human Activity Recognition for Praying" (HAR-P). The dataset includes accelerometer, linear acceleration, magnetic field, and gyroscope sensor data collected from 50 male volunteers between the ages of 18–60 while they engaged in various activities, including walking, running, walking downstairs, walking upstairs, sitting, standing, typing with keyboard, and the praying.

3.1 Data collection

Data collection was carried out with great care to ensure the integrity and accuracy of the recorded sensor data. Each

volunteer wore a smartwatch on their dominant hand and a custom mobile application (SensorBox App) was used to record the sensor readings [30].

For the “walking upstairs” and “walking downstairs” activities, volunteers were instructed to ascend and descend the steps of the 3-storey building. The data collection period for each ascent and descent took approximately 45 s. For the “prayer” activity, volunteers were asked to perform 2 rak’ahs (cycles) of prayer without any time limit. This allowed them to pray naturally and comfortably, without being restricted by a specific duration. The remaining daily activities, such as walking, running, sitting, standing, and typing, were limited to a fixed duration of 1 min each. It is important to clarify that not all participants engaged in every activity. Figure 1 shows the number of activities by users.

3.2 Data labeling

After collecting the raw sensor data, the HAR-P dataset was labeled with corresponding activity names, user IDs. The dataset is structured as follows: each entry contains the user ID, activity name, timestamp, and sensor data along the x, y, and z axes (Fig. 2).

During data collection, the mobile application saved the sensor data from each sensor (accelerometer, linear accelerometer, gyroscope and magnetometer) into different files. As each sensor device may send a varying number of data points at the same sampling rate (Table 3), it is necessary to preprocess the data to combine them into a unified format with consistent frequency ranges.

The data graph for four sensor types of a user’s two-rak’ahs prayer activity is as shown in Fig. 3.

3.3 Data alignment and frequency matching

To ensure the coherency of the sensor data and to facilitate further analysis, data alignment and frequency matching methods were applied. In this process, the timestamps were synchronized to align the data from different sensors accurately (Fig. 4). To achieve data alignment and frequency matching in the HAR-P dataset, the following steps were taken during data preprocessing:

1. *Inspecting First Data Timestamps*: The timestamps of the first data points from each sensor were examined to determine the starting points of data collection.
2. *Ordering Timestamps in Ascending Order*: The timestamps from each sensor were sorted in ascending order (from earliest to latest) to establish a chronological sequence.

3. *Identifying the Highest Timestamp*: The highest timestamp among all sensors’ first records was identified to determine the endpoint of data processing.
4. *Locating the Second Record of the Highest Timestamp*: The second record of the highest timestamp was identified to initiate the alignment process.
5. *Averaging Sensor Values for Frequency Matching*: From the first record to the highest timestamp of the second record, the sensor values (accelerometer, linear accelerometer, gyroscope, and magnetometer) were averaged to ensure that all sensors had the same frequency range.
6. *Writing Aligned Data to a New File*: The averaged sensor values, along with user ID, activity name, and timestamp, were written to a new file in the desired format. The new file now contains data from all sensors aligned in a synchronized manner.

After achieving the alignment and frequency matching, we organized the sensor data into a unified dataset structure as Table 4.

Each entry in the HAR-P dataset now consists of the following columns:

User-id: The identifier of the individual volunteer.

Activity: The name of the activity performed by the volunteer.

Timestamp: The synchronized timestamp that ensures all sensors have data at the same time intervals.

acgX, acgY, acgZ (Accelerometer): The average values of accelerometer data along the X, Y, and Z axes at the given timestamp. Accelerometer sensor, quantifies acceleration forces (m/s^2) on a device across 3 axes (x, y, z), incorporating gravity.

gyroX, gyroY, gyroZ (Gyroscope): The average values of gyroscope data along the X, Y, and Z axes at the given timestamp. Gyroscope sensor, quantifies device rotation speed (rad/s) across 3 axes (x, y, z).

magnetX, magnetY, magnetZ (Magnetic Field): The average values of magnetic field sensor data along the X, Y, and Z axes at the given timestamp. Magnetic field sensor, gauges the surrounding geomagnetic field across 3 axes (x, y, z) in microtesla (μT).

accX, accY, accZ (Linear Acceleration): The average values of linear acceleration data along the X, Y, and Z axes at the given timestamp. Linear Acceleration sensor, quantifies applied acceleration forces (m/s^2) on a device across 3 axes (x, y, z), excluding gravitational force.

However, for the purpose of this study, we transformed the original dataset with 8-class into a 2-class dataset by combining all activities except praying into a single class, which we refer to as “Non-Praying” activities (Fig. 5).

The data distribution graph after making the data set into 2-class can be seen in Fig. 6.

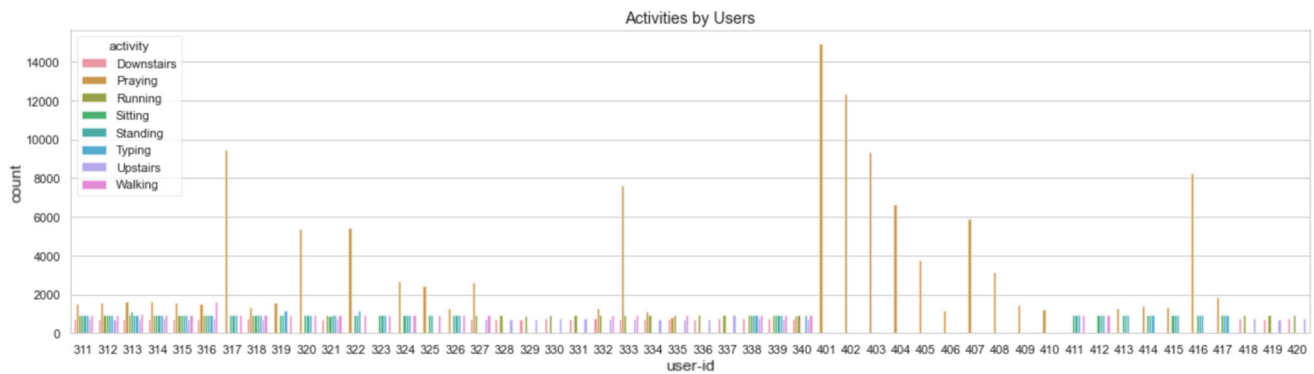


Fig. 1 Graph of activity counts by user

Fig. 2 Labeled data, datatype of HAR-P

```
311,Praying,168595864301773,9.702091,-1.5278633,0.807112
311,Praying,168595883532773,9.728394,-1.5804696,0.759288
311,Praying,168595909223773,9.723612,-1.549384,0.711464
311,Praying,168595928754773,9.728394,-1.5087335,0.74972314
311,Praying,168595948285773,9.709265,-1.4274328,0.7760264
311,Praying,168595968793773,9.5897045,-1.4824303,0.6899432
311,Praying,168595988324773,9.632747,-1.5374279,0.6947256
311,Praying,168596008832773,9.685353,-1.5565577,0.6732048
311,Praying,168596028363773,9.694918,-1.5278633,0.651684
311,Praying,168596047894773,9.726004,-1.5159072,0.69950795
```

Table 3 HAR-P dataset (with 8-class) description. Number of samples by activity and sensors

	Linear acceleration	Accelerometer	Gyroscope	Magnetic field
Downstairs	17,987–6.4%	60,955–6.28%	59,701–6.45%	17,954–6.47%
Praying	127,114–45.24%	449,985–46.35%	415,092–44.87%	124,155–44.72%
Running	23,259–8.28%	79,350–8.17%	77,008–8.32%	23,221–8.36%
Sitting	23,414–8.33%	81,362–8.38%	77,687–8.4%	23,348–8.41%
Standing	23,231–8.27%	77,409–7.97%	77,157–8.34%	23,210–8.36%
Typing	23,647–8.42%	79,317–8.17%	78,221–8.46%	23,516–8.47%
Upstairs	18,369–6.54%	61,517–6.34%	60,795–6.57%	18,325–6.6%
Walking	23,959–8.53%	80,845–8.33%	79,448–8.59%	23,924–8.62%
TOTAL	280,980–100%	970,740–100%	925,109–100%	277,653–100%

4 Methodology

The methodology employed in this study encompasses the utilization of the HAR-P dataset for salat recognition, coupled with the implementation of advanced deep learning models and cross-validation techniques. Figure 7 shows, architecture of our approach. In the segmentation phase, sensor data streams are often divided into sub-sequences called “windows” for processing. There are several windowing techniques in the literature: activity-defined windows, event-defined windows and sliding windows [31]. The most popular among them is the sliding window approach which is shifted over the stream of sensor data. The sliding window approach has two parameters; the length of the window (time_step) and the

amount of transitions between consecutive windows (step_size). Various window sizes from 0.1 s [32] to 60 s [33] have been used in previous studies. It is recommended to use a larger window size for complex activities[31].

In this study, we tried different window sizes for prayer recognition by making a pre-test. As a result, other tests were performed with a window size of 200 “time_steps” (approximately 12.9 s) and 10 “step_size” shifts (95% overlap).

After segmentation, the data should be divided into training set and testing set. When dealing with a binary classification problem with an imbalanced dataset (60%, 40%), using stratified k-fold cross-validation is a good practice. Stratified k-fold cross-validation technique ensures that each fold of the cross-validation retains the

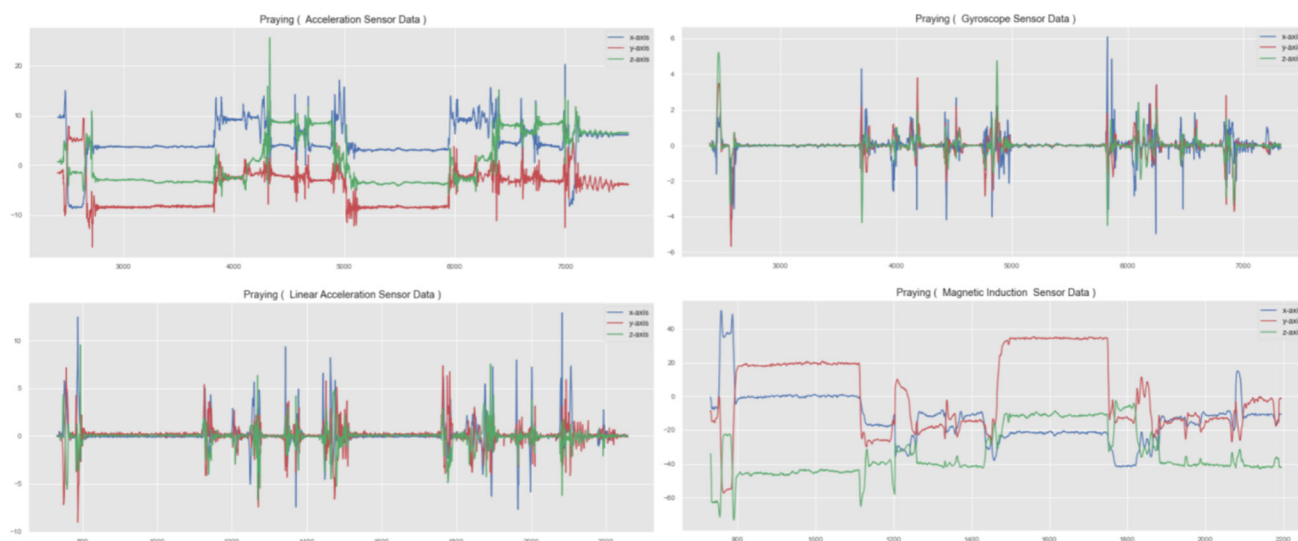


Fig. 3 Four sensor data graphs for two-rak'ahs (cycle) of prayer activity

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311,Praying,168595864301773,9.702091,-1.5278633,0.807112
311,Praying,168595883532773,9.728394,-1.5804606,0.759288
311,Praying,168595909223773,9.723612,-1.549384,0.711464
311,Praying,168595928754773,9.728394,-1.5087315,0.74972314
311,Praying,168595948285773,9.709265,-1.4274328,0.7760264
311,Praying,168595968793773,9.5897045,-1.4824303,0.6899432
311,Praying,168595988324773,9.632747,-1.5374279,0.6947256
311,Praying,168596008832773,9.685353,-1.5565577,0.6732048
311,Praying,16859602836773,9.694918,-1.5278633,0.651684
311,Praying,168596047894773,9.726004,-1.5159072,0.69950795

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311,Praying,WEAR_ACC

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311,Praying,168595991565604,-0.04031592,0.14538164,-0.037872527
311,Praying,16859601096604,0.009773556,0.053754557,-0.0036650833
311,Praying,168596030627604,0.020768806,0.01343639,0.00488678
311,Praying,168596051135604,-0.006108472,0.018325416,0.0012216945
311,Praying,16859607066604,-0.020768806,0.015882028,0.006108472
311,Praying,16859609174604,-0.014660333,0.013438639,0.013438639
311,Praying,16859610705604,0.0036650833,0.0073301666,0.018325416
311,Praying,16859613023604,0.019547112,0.00243389,0.019547112
311,Praying,168596149768604,0.015882028,-0.0036650833,0.014660333
311,Praying,16859616929604,-0.006108472,0.00243389,0.01703722

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311,Praying,WEAR_GYRO

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311,Praying,168596179040967,-1.02,-8.5199995,-34.68
311,Praying,168596243493967,-4.3199997,-10.559999,-49.079998
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311,Praying,168596372399967,-7.44,-14.88,-62.34
311,Praying,168596436852967,-7.3199997,-15.059999,-63.12
311,Praying,168596501305967,-6.6,-14.759999,-63.18
311,Praying,168596565758967,-6.42,-14.46,-62.879997
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311,Praying,WEAR_MAGNET

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311,Praying,168596451218488,-0.09377872,0.01651968,-0.0056345835
311,Praying,168596515671488,0.4795154,-0.08010662,0.041496485
311,Praying,168596580124488,0.04921248,-0.01755509,0.014177021
311,Praying,168596644577488,-0.02724319,0.0019250254,-0.004077001
311,Praying,168596709030488,-0.13315919,0.012144858,-0.4656945
311,Praying,168596773483488,0.06179202,0.1429594,-0.5064772

```

311,Praying,WEAR_ACC

Data Alignment and Frequency Matching

user-id, activity, timestamp, acgX, acgY, acgZ, gyroX, gyroY, gyroZ, magnetX, magnetY, magnetZ, accX, accY, accZ

```

311,Praying,168596193406488,9.70518561764706,-1.49466772352941,0.71849693,-0.00377614642727273,0.0244338891181818,0.00744122992727273,-1.02,-8.5199995,-34.68,-0.08126403,0.0063581434,-0.01181867
311,Praying,168596257859488,9.7076705,-1.45851843333333,0.651684,-0.0456099266666667,0.001628926,0.0211760373333333,-4.3199997,-10.559999,-49.079998,0.017901646,-0.0001468319,0.00060987717
311,Praying,16859632212488,10.1588103333333,1.64501206666667,0.77921468,-0.0553834816666667,-0.00366508333333333,0.0101807866666667,-7.2,-14.34,-62.52,0.0035863146,0.0012043748,-0.0024948954
311,Praying,168596451218488,9.594156666667,-1.46170671,0.04276,0.209211725,-0.0574196385,-0.2162399075,-7.3199997,-15.059999,-63.12,-0.09377872,0.01651968,-0.0056345835
311,Praying,168596515671488,9.51796833333333,-1.249687,0.639728,-0.0476460843333333,0.0769667466666667,-0.0655642766666667,-6.6,-14.759999,-63.18,0.4795154,-0.08010662,0.041496485
311,Praying,168596580124488,9.7100615,-1.26164296666667,0.1925756,-0.209316993333333,0.20076512,0.12257668,-6.42,-14.46,-62.879997,0.04921248,-0.01755509,0.014177021
311,Praying,168596644577488,9.71344875,-1.020331025,0.50761417,-0.3233418,0.173887846666667,0.166557663333333,-7.14,-14.7,-62.879997,-0.02724319,0.0019250254,-0.004077001
311,Praying,168596709030488,9.59767533333333,-0.984263686666667,0.779214666666667,-0.1163663925,-0.00549762575,0.016798301,-7.98,-15.059999,-62.94,-0.13315919,0.012144858,-0.4656945
311,Praying,168596773483488,9.820854,-1.29751093333333,0.666031,1.66666667,-0.102215096666667,-0.106694643333333,-0.08633307,-7.68,-15.599999,-62.64,0.06179202,0.1429594,-0.5064772

```

Fig. 4 Four sensor files aligning as a file with frequency matching

Table 4 Data structure of HAR-P dataset

User-id	Activity	Timestamp	acgX	acgY	acgZ	gyroX	gyroY	gyroZ	magnetX	magnetY	magnetZ	accX	accY	accZ
---------	----------	-----------	------	------	------	-------	-------	-------	---------	---------	---------	------	------	------

same distribution of activities as the original dataset, thereby mitigating potential bias that could arise from imbalanced class representation. Model training and

evaluations were carried out after preliminary preparation. The training and evaluation stages are explained in the following sections.

4.1 Deep learning models

To unravel the intricate patterns associated with salat recognition, we employed three deep learning models known for their efficacy in activity recognition:

4.1.1 LSTM (Long short-term memory)

LSTM, a type of recurrent neural network (RNN), is designed to capture temporal dependencies in sequential data. LSTM's ability to retain information over extended sequences is particularly suited to capturing the dynamic transitions present in activities like salat.

Figure 8 shows LSTM model architecture which we use. We used the LSTM model with the parameters in Table 5.

4.1.2 CNN-LSTM (Convolutional neural network—long short-term memory)

Combining the strengths of CNNs in spatial feature extraction with the temporal memory capabilities of LSTMs, the CNN-LSTM architecture aims to effectively capture both spatial and temporal features present in sensor data.

Figure 9 shows CNN-LSTM model architecture which we use. We used the CNN-LSTM model with the parameters in Table 6.

4.1.3 ConvLSTM (Convolutional LSTM)

The ConvLSTM model extends traditional LSTMs by incorporating convolutional operations within the LSTM cell. This architecture is particularly well-suited to spatio-temporal data like sensor sequences, making it an ideal candidate for recognizing salat activities where both the spatial arrangement of body positions and the temporal flow of movements are critical.

Figure 10 shows ConvLSTM model architecture which we use. We used the ConvLSTM model with the parameters in Table 7.

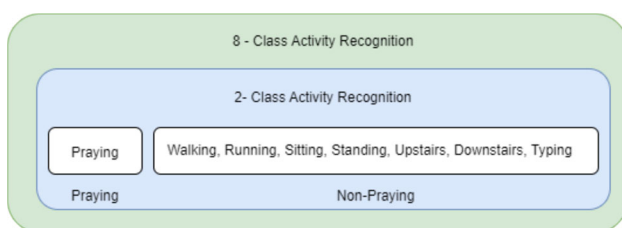


Fig. 5 Dataset attributes

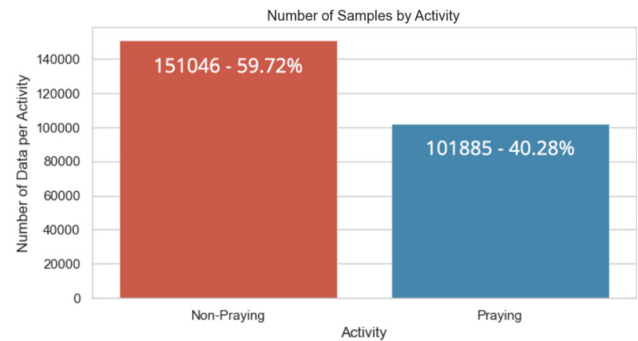


Fig. 6 HAR-P dataset (with 2-class) description

4.2 Experimental setup

We designed a series of experiments to comprehensively evaluate sensor group combinations. We explored all possible combinations of sensor groups using the combinations in Table 8. The experiments were carried out in a Python environment with keras 2.7.0, scikit-learn 1.2.2, pandas 1.3.4, numpy 1.25.2, python 3.9.17 libraries.

5 Results

Salat recognition results on the HAR-P dataset for three deep learning models are presented in this section. We reported and compared the salat recognition performance of the models; using the evaluation metrics such as accuracy, precision, recall and F1-score. We employed a technique known as ‘early stopping’ during the training phase of neural network. Early stopping involves monitoring a specified metric, such as validation loss or accuracy, during the training process. The training is terminated when this metric ceases to improve, indicating that further iterations would likely lead to overfitting. We used “loss” metric with patience = 8, epoch = 150 parameters for early stopping.

For the training of models, we systematically explored different batch sizes to determine the optimal setting. Prior to the main experiments, a preliminary batch size analysis was conducted. During this phase, various batch sizes were tested, ranging from 32 to 256. Based on the results of the pre-test, we found that a batch size of 128 yielded the best performance in terms of validation accuracy. As a result, we selected a batch size of 128 for all subsequent experiments and tests.

5.1 LSTM model test results

Table 9 represents the test results of the LSTM model with fivefold cross validation. The explanation of the sensor group expressions in the first column of Table 9 is given in



Fig. 7 Architecture of our approach

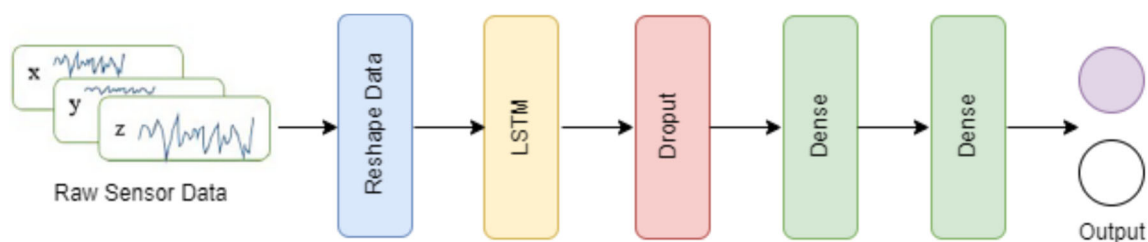


Fig. 8 LSTM model architecture

Table 5 The dimensional structure of the LSTM model

Layer name	Parameter settings
LSTM	lstm_neurons: 100
Dropout	dropout: 0.5
Dense	dense_neurons: 100; activation: relu
Dense	num_classes: 2; activation: sigmoid

Table 6 The dimensional structure of the CNN- LSTM model

Layer name	Parameter settings
Conv1D	Filter: 64; kernel_size: 3; activation: relu
Conv1D	Filter: 64; kernel_size: 3; activation: relu
Dropout	Dropout: 0.5
MaxPooling1D	Pool_size: 2
Flatten	
LSTM	lstm_neurons: 100
Dropout	Dropout: 0.5
Dense	Dense_neurons: 100; activation: relu
Dense	Num_classes: 2; activation: sigmoid

Table 8. The average accuracy value of the fivefold of the LSTM model is shown in Table 10. In order to see the effect of the number of sensors on the classification success, Table 10 is grouped according to the number of sensors.



Fig. 9 CNN-LSTM model architecture

According to the information given in Table 10, apart from the addition of the linear acceleration sensor (G8) to the gyroscope sensor (G3), the use of two sensors instead of one increased the success of salat recognition classification. Adding the gyro sensor (G11) to the acc-acg dual sensor group (G5) slightly increased the classification success. The addition of the magnet sensor (G14) to the acc-gyro dual sensor group (G8) again increased the classification success slightly. However, adding an additional third sensor (G12 and G13) to the other dual sensor groups (G6, G7, G9 and G10) reduced the success. Using the quadruple sensor group (G15) instead of the dual sensor group (G6, G7, G8, G9, 10) increased the classification success. There was no change only in the acc-acg sensor group (G5). Using the quadruple sensor group (G15) instead of the triple sensor group (G12, G13, G14) increased the classification success except for the acc-acg-gyro sensor group (G11). The information in Table 10 has been converted into a line graph to make it easier to read. The line graph of the average accuracy values of the LSTM model can be seen in Fig. 11. The highest accuracy rate (95.4%) for the LSTM model was achieved with linear acceleration—acceleration—gyroscope (G11) sensor combinations. The lowest success rate 86.0% for the LSTM model was achieved with the singular use of the linear acceleration sensor.

5.2 CNN- LSTM model test results

Table 11 represents the test results of the CNN-LSTM model with fivefold cross validation. The average accuracy value of the fivefold of the CNN-LSTM model is shown in Table 12.

According to the information given in Table 12, using two sensors instead of one increased the success of salat recognition and classification. The use of a triple sensor group instead of a double sensor group for the CNN-LSTM model increased the classification success, except for the addition of the magnet sensor (G12) to the acc-acg sensor group (G5). For the CNN-LSTM model, the use of quadruple sensor group (G15) instead of a triple sensor group increased the success in some (G12, G13) and decreased the success in others (G11, G14).

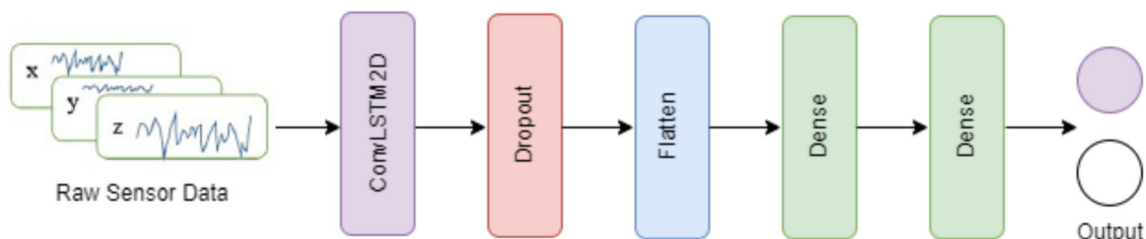


Fig. 10 ConvLSTM model architecture

Table 7 The dimensional structure of the ConvLSTM model

Layer name	Parameter settings
ConvLSTM2D	Filter: 64, kernel_size: (1,3); activation: relu
Dropout	Dropout: 0.5
Flatten	
Dense	Dense_neurons: 100; activation: relu
Dense	Num_classes: 2; activation: sigmoid

Table 8 Sensor group combinations

Group Name	Sensor(s)
G1	acg
G2	acc
G3	gyro
G4	magnet
G5	acg and acc
G6	acg and gyro
G7	acg and magnet
G8	acc and gyro
G9	acc and magnet
G10	gyro and magnet
G11	acg, acc and gyro
G12	acg, acc and magnet
G13	acg, gyro and magnet
G14	acc, gyro and magnet
G15	acg, acc, gyro and magnet

Sensor(s) acg (accelerometer), acc (linear acceleration), gyro (gyroscope), magnet (magnetic field)

The information in Table 12 has been converted into a line graph to make it easier to read. The line graph of the average accuracy values of the CNN-LSTM model can be seen in Fig. 12. The highest accuracy rate (96.4%) for the CNN-LSTM model was achieved with linear acceleration—acceleration—gyroscope (G11) sensor combinations. The lowest success rate (92.5%) for the CNN-LSTM model was achieved with the single use of the magnet sensor.

Table 9 Sensor group combination test results for LSTM model

LSTM Model Sensor Groups	Fold 1				Fold 2				Fold 3			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
G1	0.953	0.982	0.899	0.939	0.934	0.948	0.884	0.915	0.931	0.911	0.920	0.915
G2	0.875	0.914	0.761	0.831	0.857	0.875	0.752	0.809	0.817	0.853	0.659	0.744
G3	0.889	0.973	0.747	0.845	0.890	0.958	0.761	0.848	0.889	0.963	0.755	0.846
G4	0.901	0.899	0.850	0.874	0.920	0.936	0.861	0.897	0.931	0.930	0.896	0.913
G5	0.937	0.933	0.909	0.920	0.954	0.979	0.904	0.940	0.951	0.969	0.907	0.937
G6	0.958	0.978	0.917	0.946	0.955	0.976	0.914	0.943	0.916	0.994	0.795	0.884
G7	0.927	0.921	0.895	0.908	0.945	0.952	0.910	0.930	0.940	0.939	0.911	0.925
G8	0.911	0.933	0.839	0.884	0.875	0.898	0.778	0.834	0.788	0.854	0.573	0.685
G9	0.952	0.936	0.947	0.941	0.932	0.949	0.879	0.913	0.949	0.963	0.909	0.935
G10	0.949	0.939	0.933	0.936	0.939	0.921	0.927	0.924	0.953	0.933	0.950	0.942
G11	0.961	0.984	0.919	0.950	0.950	0.965	0.909	0.936	0.943	0.947	0.910	0.928
G12	0.938	0.926	0.920	0.923	0.927	0.904	0.915	0.910	0.938	0.940	0.905	0.922
G13	0.935	0.933	0.902	0.917	0.908	0.916	0.851	0.882	0.923	0.915	0.891	0.903
G14	0.811	0.823	0.678	0.743	0.940	0.929	0.922	0.926	0.898	0.900	0.841	0.870
G15	0.942	0.908	0.954	0.930	0.941	0.941	0.911	0.926	0.960	0.964	0.937	0.950
LSTM Model Sensor Groups	Fold 4				Fold 5							
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score				
G1	0.902	0.902	0.850	0.875	0.949	0.970	0.901	0.934				
G2	0.880	0.889	0.804	0.844	0.869	0.826	0.855	0.840				
G3	0.906	0.893	0.872	0.883	0.909	0.885	0.891	0.888				
G4	0.915	0.893	0.898	0.895	0.930	0.921	0.903	0.912				
G5	0.959	0.971	0.925	0.948	0.952	0.967	0.912	0.939				
G6	0.961	0.991	0.912	0.950	0.954	0.966	0.918	0.941				
G7	0.932	0.932	0.897	0.914	0.939	0.956	0.890	0.922				
G8	0.882	0.915	0.779	0.842	0.959	0.965	0.932	0.948				
G9	0.949	0.955	0.917	0.935	0.945	0.940	0.922	0.931				
G10	0.947	0.937	0.932	0.934	0.952	0.956	0.923	0.939				
G11	0.958	0.972	0.923	0.947	0.957	0.978	0.914	0.945				
G12	0.889	0.844	0.888	0.866	0.944	0.960	0.898	0.928				
G13	0.921	0.925	0.874	0.899	0.927	0.911	0.907	0.909				
G14	0.907	0.909	0.855	0.881	0.924	0.907	0.904	0.905				
G15	0.955	0.974	0.912	0.942	0.956	0.973	0.916	0.944				

5.3 Conv LSTM model test results

Table 13 represents the test results of the ConvLSTM model with fivefold cross validation. The average accuracy value of the fivefold of the ConvLSTM model is shown in Table 14.

According to the information given in Table 14, a general statement could not be reached about whether using two sensors instead of one sensor for the ConvLSTM model will increase or decrease the success. While adding

a magnet sensor to the linear acceleration sensor reduced the success, adding the gyroscope and acceleration sensor separately increased the success. While adding the linear acceleration sensor to the acceleration sensor did not change the success, adding the magnet and gyroscope sensor separately decreased the success. While adding a magnet sensor to the gyroscope sensor reduced the success, adding the linear acceleration and acceleration sensors separately increased the success. While adding the acceleration and gyroscope sensors to the Magnet sensor

Table 10 LSTM model test results (according to sensor type numbers)

LSTM Model Test Results					
Singel Sensor		Dual Sensor		Triple Sensor	
				Quadruple Sensor	
		acc-acg	0.951		
acc	0.860	acc-gyro	0.883	acc-acg-gyro	0.954
acg	0.934	acc-magnet	0.945	acc-gyro-magnet	0.923
gyro	0.897	acg-gyro	0.949	acc-acg-gyro-magnet	
magnet	0.919	acg-magnet	0.937	acc-acg-magnet	0.927
		gyro-magnet	0.948	acg-gyro-magnet	0.896

increased the success, adding the linear acceleration sensor separately decreased the success.

For the ConvLSTM model, success has increased in the triple sensor groups (G11, G12, G13) obtained by separately adding the acceleration sensor to the double sensor groups (G8, G9, G10). For the ConvLSTM model, adding the magnet sensor separately to the dual sensor groups G5 and G6 increased the success, while there was no change in G8. For the ConvLSTM model, adding the gyroscope sensor separately to the dual sensor groups G7 and G9 increased the success rate, while it decreased the performance slightly in G5. For the ConvLSTM model, the addition of the linear acceleration sensor to the dual sensor groups G7 and G10 increased the success, while there was no change in G6.

For the ConvLSTM model, using a quadruple sensor group (G15) instead of a triple sensor group (G11, G13 and G14) increased the success, except for the acc-acg-magnet sensor group (G12).

The information in Table 14 has been converted into a line graph to make it easier to read. The line graph of the average accuracy values of the ConvLSTM model can be seen in Fig. 13. The highest accuracy rate (96.1%) for the ConvLSTM model was achieved with linear acceleration—acceleration—magnetic field sensor (G12) combinations.

The lowest success rate for the ConvLSTM model, 81.2%, was obtained by using linear acceleration and magnetic field sensor together (G9).

Figure 14 summarizes the obtained results from the three models with all sensor group combinations. The highest accuracy rate from the single sensor group (95.7%) was achieved with the ConvLSTM method with the acceleration sensor. Among the single sensor groups for all three models, the acceleration sensor achieved the highest success. The highest accuracy rate of 95.9% from dual sensor groups was achieved with the CNN-LSTM method. Among the dual sensor groups, the highest success was achieved for all three models with acceleration and linear acceleration sensor combinations. Among the triple sensor groups, the highest accuracy rate of 96.4% was achieved with the CNN-LSTM method. In the quadruple sensor group, the highest accuracy rate of 96.0% was again achieved with the CNN-LSTM method.

In our prayer recognition study, the highest accuracy rate of 96.4% was achieved with the CNN-LSTM method using the acc-acg-gyro (G11) sensor group. The lowest accuracy rate of 81.2% was achieved with the ConvLSTM method using the acc-magnet (G9) sensor group. The highest success, both in the average of 15 sensor groups (95.4%) and in the overall result (96.4%), was achieved

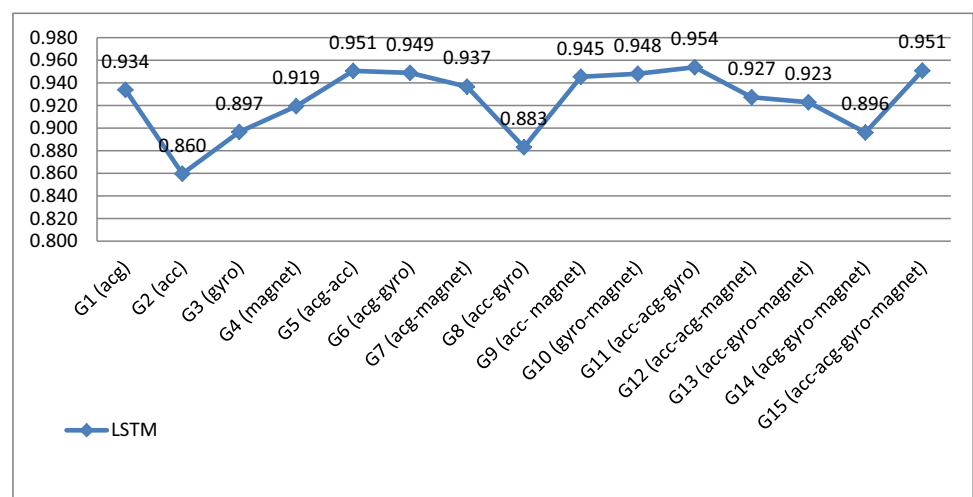
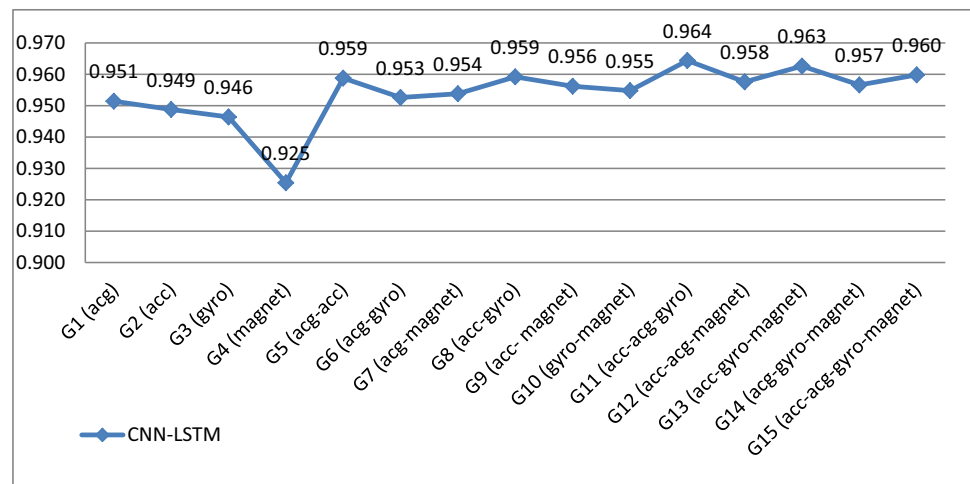
Fig. 11 LSTM model test results chart

Table 11 Sensor group combination test results for CNN-LSTM model

CNN-LSTM Model Sensor Groups	Fold 1				Fold 2				Fold 3			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
G1	0.955	0.992	0.895	0.941	0.942	0.986	0.869	0.924	0.953	0.986	0.896	0.939
G2	0.952	0.972	0.907	0.938	0.948	0.959	0.911	0.934	0.946	0.959	0.903	0.931
G3	0.958	0.973	0.923	0.947	0.942	0.932	0.923	0.928	0.948	0.965	0.902	0.933
G4	0.924	0.929	0.878	0.903	0.934	0.916	0.921	0.919	0.925	0.941	0.869	0.904
G5	0.962	0.990	0.916	0.952	0.952	0.962	0.917	0.939	0.957	0.984	0.907	0.944
G6	0.955	0.990	0.896	0.941	0.951	0.979	0.898	0.937	0.952	0.988	0.892	0.937
G7	0.952	0.997	0.885	0.937	0.952	0.988	0.893	0.938	0.953	0.982	0.901	0.940
G8	0.963	0.982	0.925	0.953	0.959	0.975	0.921	0.947	0.956	0.958	0.933	0.945
G9	0.958	0.976	0.918	0.946	0.952	0.983	0.896	0.938	0.955	0.980	0.906	0.942
G10	0.948	0.904	0.974	0.937	0.953	0.931	0.955	0.943	0.955	0.931	0.958	0.944
G11	0.965	0.994	0.919	0.955	0.963	0.994	0.913	0.952	0.962	0.985	0.919	0.951
G12	0.955	0.996	0.891	0.940	0.960	0.990	0.911	0.949	0.947	0.985	0.881	0.930
G13	0.964	0.978	0.932	0.955	0.956	0.965	0.921	0.944	0.961	0.976	0.927	0.951
G14	0.957	0.987	0.906	0.945	0.955	0.991	0.896	0.941	0.959	0.980	0.910	0.947
G15	0.963	0.986	0.921	0.953	0.957	0.987	0.906	0.945	0.957	0.984	0.910	0.945
CNN-LSTM Model Sensor Groups	Fold 4				Fold 5							
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score				
G1	0.956	0.985	0.905	0.943	0.951	0.982	0.895	0.936				
G2	0.949	0.955	0.917	0.936	0.949	0.970	0.902	0.935				
G3	0.941	0.926	0.928	0.927	0.943	0.955	0.901	0.928				
G4	0.930	0.935	0.889	0.911	0.914	0.881	0.909	0.895				
G5	0.964	0.984	0.925	0.954	0.959	0.985	0.912	0.947				
G6	0.957	0.983	0.909	0.944	0.948	0.990	0.879	0.931				
G7	0.958	0.995	0.901	0.946	0.954	0.984	0.901	0.941				
G8	0.958	0.956	0.940	0.948	0.960	0.962	0.937	0.950				
G9	0.957	0.969	0.923	0.945	0.959	0.979	0.918	0.947				
G10	0.958	0.948	0.949	0.948	0.960	0.945	0.956	0.950				
G11	0.969	0.988	0.933	0.960	0.963	0.990	0.917	0.952				
G12	0.963	0.983	0.925	0.953	0.963	0.991	0.918	0.953				
G13	0.967	0.979	0.938	0.958	0.965	0.979	0.934	0.956				
G14	0.956	0.995	0.895	0.942	0.956	0.990	0.901	0.943				
G15	0.963	0.985	0.923	0.953	0.959	0.987	0.910	0.947				

Table 12 CNN-LSTM model test results (according to sensor type numbers)

CNN-LSTM model test results									
Singel sensor		Dual sensor		Triple sensor		Quadruple sensor			
		acc-acg	0.959						
acc	0.949	acc-gyro	0.959	acc-acg-gyro	0.964				
acg	0.951	acc-magnet	0.956	acc-gyro-magnet	0.963				
gyro	0.946	acg-gyro	0.953			acc-acg-gyro-magnet	0.96		
magnet	0.925	acg-magnet	0.954	acc-acg-magnet	0.958				
		gyro-magnet	0.955	acg-gyro-magnet	0.957				

Fig. 12 CNN-LSTM model test results chart

with the CNN-LSTM method. The sensor groups in which the three models achieved the highest average success were: G5, G6, G11 and G15.

5.4 Discussion

In the present study, the sensor data, acg (accelerometer), acc (linear acceleration), gyro (gyroscope), magnet (magnetic field), have been collected to determine human activity recognition. The dataset have been created by using single, dual, triple and quadruple sensor configurations. The obtained data are used for human activity recognition by using the three deep learning models, LSTM, CNN-LSTM, and ConvLSTM. For LSTM model, Accuracy scores range from 0.860 to 0.954, indicating the proportion of correctly classified instances out of the total instances. For CNN-LSTM model, accuracy scores range from 0.946 to 0.964, indicating the proportion of correctly classified instances out of the total instances. For ConvLSTM, Accuracy scores range from 0.812 to 0.961, representing the ratio of correctly predicted instances to the total instances. When we consider all the models and all sensor configurations, as the number of sensors increases, the performance of the model also increases. However, the better results are obtained by using triple sensor configurations in accordance with Tables 10, 12 and 14. For the deep learning models and sensor configuration, the highest accuracy for single sensor configuration comes from acg. In the other sensor configurations, while acg is in the configuration, the better results are obtained. The better results in all the comparisons are obtained by using the triple sensor configuration (acc-acg-gyro) and CNN-LSTM model. When we compare the models together:

LSTM Model: The LSTM model achieves accuracy scores ranging from 0.860 to 0.954. It generally performs

well but may have slightly lower accuracy compared to the other models.

CNN-LSTM Model: The CNN-LSTM model demonstrates higher accuracy scores compared to the LSTM model, ranging from 0.925 to 0.964. It consistently achieves higher scores across different sensor configurations.

ConvLSTM Model: The ConvLSTM model also shows strong performance with accuracy scores ranging from 0.892 to 0.961. It achieves competitive results, particularly in configurations with multiple sensors.

The CNN-LSTM and ConvLSTM models, which incorporate convolutional layers, generally outperform the traditional LSTM model across various sensor configurations. This suggests that leveraging spatial information through convolutional layers improves performance. The choice of sensor combination significantly influences model performance. Certain combinations, such as acc-acg-gyro, consistently yield better accuracy scores. While increasing the number of sensors in the configuration tends to improve accuracy, it may also introduce complexity, synchronization problems, etc.

6 Conclusions and future work

In this study, we aimed to develop an effective human activity recognition system for detecting the specific activity of the praying within a diverse set of daily activities. We collected sensor data from smartwatches worn by 50 male volunteers performing various activities, including walking, running, sitting, standing, walking upstairs, walking downstairs, typing, and praying. We employed accelerometer, linear accelerometer, magnetometer, and gyroscope sensors to create the HAR-P (Human Activity

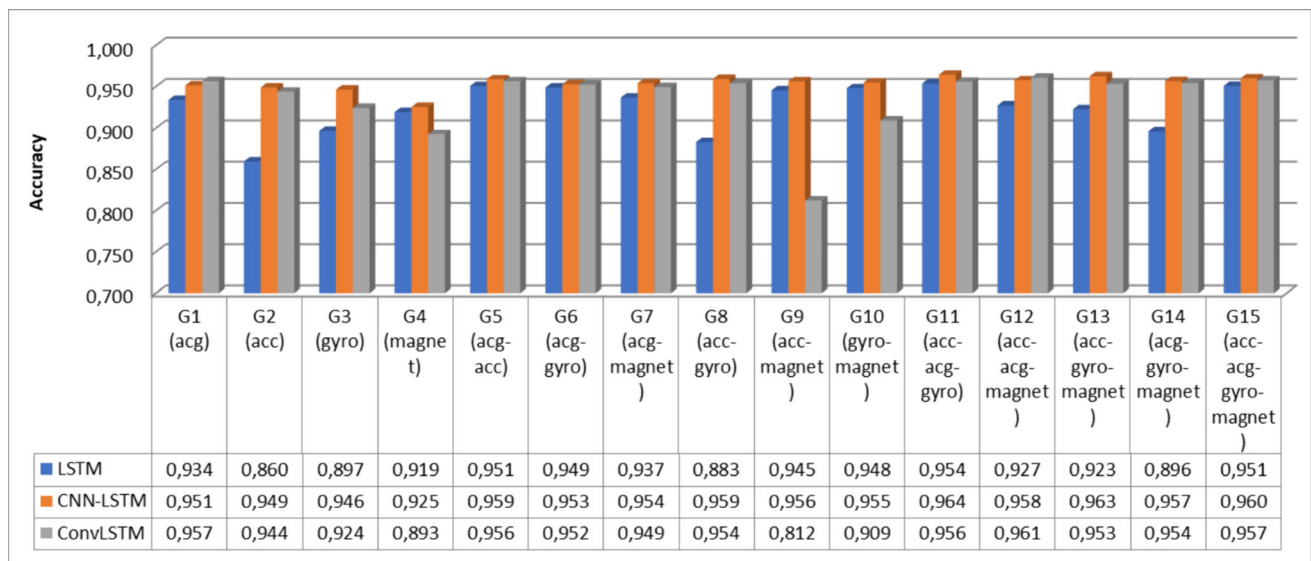
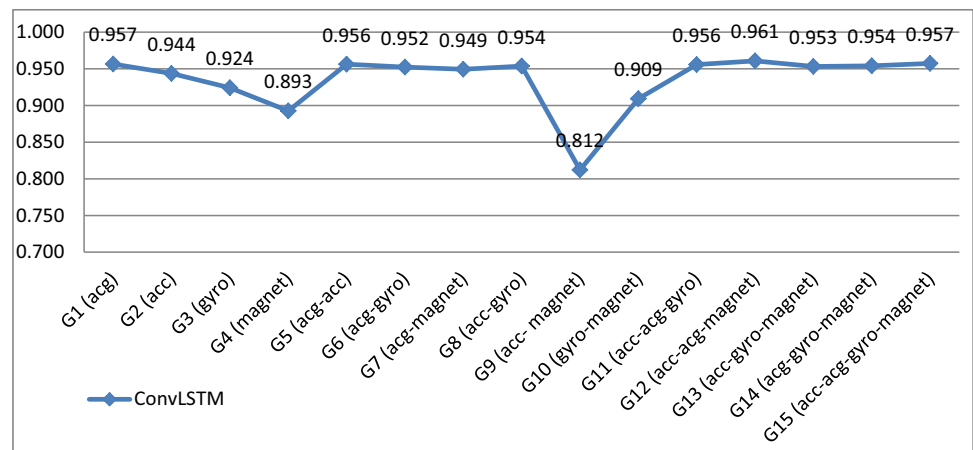
Table 13 Sensor group combination test results for ConvLSTM model

ConvLSTM Model Sensor Groups	Fold 1				Fold 2				Fold 3			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
G1	0.957	0.978	0.914	0.945	0.957	0.982	0.910	0.945	0.954	0.987	0.899	0.941
G2	0.944	0.954	0.905	0.929	0.937	0.953	0.888	0.919	0.944	0.960	0.898	0.928
G3	0.901	0.901	0.849	0.874	0.939	0.944	0.903	0.923	0.931	0.901	0.931	0.916
G4	0.924	0.919	0.892	0.905	0.846	0.863	0.737	0.795	0.923	0.902	0.907	0.905
G5	0.958	0.984	0.912	0.946	0.955	0.986	0.901	0.942	0.955	0.979	0.908	0.942
G6	0.956	0.974	0.915	0.943	0.950	0.976	0.899	0.936	0.956	0.979	0.910	0.943
G7	0.956	0.947	0.945	0.946	0.932	0.936	0.894	0.914	0.937	0.922	0.922	0.922
G8	0.957	0.970	0.922	0.945	0.944	0.947	0.913	0.929	0.951	0.965	0.912	0.938
G9	0.880	0.830	0.882	0.855	0.915	0.892	0.897	0.895	0.845	0.822	0.784	0.803
G10	0.929	0.914	0.909	0.911	0.891	0.842	0.897	0.869	0.910	0.868	0.917	0.892
G11	0.956	0.979	0.912	0.944	0.957	0.994	0.899	0.944	0.953	0.967	0.915	0.940
G12	0.963	0.970	0.939	0.954	0.958	0.956	0.940	0.948	0.958	0.981	0.914	0.946
G13	0.950	0.931	0.945	0.938	0.955	0.964	0.924	0.943	0.953	0.951	0.930	0.941
G14	0.949	0.929	0.945	0.937	0.957	0.977	0.916	0.946	0.966	0.978	0.937	0.957
G15	0.948	0.949	0.919	0.934	0.962	0.973	0.930	0.951	0.959	0.965	0.932	0.948

ConvLSTM Model Sensor Groups	Fold 4				Fold 5			
	Accuracy	Precision	Recall	F1-Score	Accuracy	Precision	Recall	F1-Score
G1	0.958	0.986	0.909	0.946	0.957	0.975	0.917	0.945
G2	0.945	0.959	0.903	0.930	0.949	0.941	0.932	0.936
G3	0.934	0.968	0.866	0.914	0.916	0.898	0.892	0.895
G4	0.921	0.921	0.881	0.900	0.849	0.780	0.872	0.823
G5	0.957	0.987	0.905	0.944	0.957	0.984	0.908	0.945
G6	0.947	0.980	0.885	0.930	0.953	0.983	0.898	0.939
G7	0.960	0.966	0.935	0.950	0.962	0.953	0.952	0.953
G8	0.956	0.960	0.929	0.944	0.961	0.973	0.930	0.951
G9	0.784	0.769	0.665	0.713	0.637	0.802	0.131	0.225
G10	0.897	0.859	0.891	0.875	0.919	0.895	0.904	0.900
G11	0.960	0.973	0.925	0.949	0.953	0.958	0.924	0.941
G12	0.963	0.977	0.929	0.953	0.962	0.975	0.930	0.952
G13	0.953	0.954	0.928	0.941	0.955	0.951	0.937	0.944
G14	0.951	0.929	0.949	0.939	0.948	0.944	0.927	0.935
G15	0.957	0.961	0.930	0.945	0.961	0.958	0.945	0.951

Table 14 ConvLSTM model test results (according to sensor type numbers)

<i>CNN-LSTM model test results</i>							
Singel sensor		Dual sensor		Triple sensor		Quadruple sensor	
		acc-acg	0.956				
acc	0.941	acc-gyro	0.953	acc-acg-gyro	0.955		
acg	0.956	acc-magnet	0.812	acc-gyro-magnet	0.953		
gyro	0.924	acg-gyro	0.955			acc-acg-gyro-magnet	0.957
magnet	0.892	acg-magnet	0.949	acc-acg-magnet	0.961		
		gyro-magnet	0.909	acg-gyro-magnet	0.954		

Fig. 13 ConvLSTM model test results chart**Fig. 14** Graph of all test results

Recognition for Praying) dataset, focusing on recognizing the salat.

During the dataset creation process, data alignment and frequency synchronization techniques were employed to merge device data with different hardware features but collected at the same sampling frequency under the same timestamp. Tests were conducted with fifteen different combinations of four different sensors to gauge the impact of sensor combinations. The results indicate that while individual sensor data can achieve high accuracy rates, incorporating multiple diverse sensor data enhances performance. The selection of sensor combinations significantly influences the success of the prayer recognition. When working with a single sensor type, preference is given to the accelerometer sensor; however, utilizing commonly available accelerometers and gyroscopes together can improve accuracy. Although the highest accuracy is achieved with the simultaneous use of three sensors,

consideration should be given to the possibility of the absence of the magnetic field sensor in every device.

Furthermore, the performances of state-of-the-art deep learning algorithms such as LSTM, CNN-LSTM, and ConvLSTM were compared using the newly created dataset. To evaluate algorithm performance, the same experimental environment was established. The CNN-LSTM hybrid method surpassed other techniques, achieving a highest accuracy rate of 96.4%. While all three algorithms were assessed in the same environment, it was noted that employing different frame sizes during data segmentation could lead to higher accuracy rates. Additionally, considering the application of the Transformer model in time series analysis since its introduction in 2017, its implementation is targeted in future stages of this study.

Subsequent to this research, a mobile application, adapted to work on smartwatches, based on the developed prayer recognition model is planned for development. This

application will provide reminders if prayers are missed and facilitate making up missed prayers if not performed on time.

Despite collecting data from various age groups, data collection was limited to male individuals adhering to the Hanafi school of thought. Differences exist in prayer practices based on gender and school of thought within Islam. Therefore, future studies are envisioned to encompass females and individuals from other schools of thought.

Funding Open access funding provided by the Scientific and Technological Research Council of Türkiye (TÜBİTAK). This study was funded by Konya Technical University Scientific Research Projects Coordination Office with project number 231113013.

Data availability The data supporting the findings of this study are not currently available in a publicly accessible repository due to manuscript pending acceptance and ongoing analysis.

Declarations

Conflict of interest Author Yasin VURGUN declares that he has no conflict of interest. Author Mustafa Servet Kiran declares that he has no conflict of interest.

Informed consent Informed consent is obtained from all individual participants included in the study.

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